

Time Value of Money (TVM) — Detailed Structured Notes

1 Fundamentals of Time Value of Money (TVM)

In finance, four fundamental variables exist:

- Interest rate (i)
- Time (n)
- Present Value (PV)
- Future Value (FV)

Knowing any three allows calculation of the fourth.

1.1 Common Applications

- Determining how long an investment takes to reach a target value.
- Determining the interest rate required for growth to a desired amount.

2 Course Methodology

- Financial calculators and Excel are not used.
- Students must rely on:
 - Mathematical formulas
 - Scientific calculators
 - Natural logarithms ($\ln$) to solve for exponents.

3 Example 1: Doubling an Investment

Question

How many years will it take for \$1,000 to become \$2,000 at an annual interest rate of 6%? Also determine the time required if the interest rate is 10%.

Formula

$$FV = PV(1 + i)^n$$

Solution at 6%

$$2000 = 1000(1.06)^n$$

$$n = \frac{\ln(2)}{\ln(1.06)} \approx 11.9 \text{ years}$$

Solution at 10%

$$n = \frac{\ln(2)}{\ln(1.10)} \approx 7.27 \text{ years}$$

4 Earnings Per Share (EPS)

Definition

$$EPS = \frac{\text{Net Income}}{\text{Outstanding Common Shares}}$$

Interpretation

EPS represents the profit allocated to each issued share and is a key measure of company performance.

5 Annuities

5.1 Types of Annuities

- **Ordinary Annuity:** Payments occur at the **end** of each period.
- **Annuity Due:** Payments occur at the **beginning** of each period.

5.2 Key Insight

Annuity Due has higher PV and FV because each payment earns interest for one extra period.

$$\text{Value of Annuity Due} = \text{Ordinary Annuity Value} \times (1 + i)$$

6 Example 2: Retirement Savings

Question

An individual saves \$5 per day, equivalent to \$1,825 per year, from age 18 to 65 (47 years). If the annual return is 8% and deposits are made at the end of each year, what will be the future value at retirement?

Classification

Payments occur at the end of each year $\Rightarrow$ Ordinary Annuity.

Formula

$$FV = PMT \times \frac{(1+i)^n - 1}{i}$$

Calculation

$$FV = 1825 \times \frac{(1.08)^{47} - 1}{0.08} \approx 826,542.86$$

7 Example 3: Lump Sum vs. Annuity Choice

Question

You must choose between:

1. Receiving \$15 million today.
2. Receiving \$1 million per year for 30 years at a 6% discount rate.

Which option is financially better?

Present Value of Annuity

$$PV = PMT \times \frac{1 - (1+i)^{-n}}{i}$$

$$PV = 1,000,000 \times \frac{1 - (1.06)^{-30}}{0.06} \approx 13.76 \text{ million}$$

Decision

$$15 \text{ million} > 13.76 \text{ million}$$

Therefore, the lump sum is the better financial choice.

8 Example 4: Time to Reach \$220,000

Question

An account currently contains \$33,556.25. An additional \$5,000 is deposited at the end of every year. If the interest rate is 12%, how many years will it take for the account balance to reach \$220,000?

Combined Future Value Equation

$$220,000 = 33,556.25(1.12)^n + 5,000 \left(\frac{(1.12)^n - 1}{0.12} \right)$$

Solve for n using logarithmic or numerical methods.

9 Professional Insights

- Finance and Accounting fields offer strong career opportunities.
- Programming is useful, but **financial domain knowledge** is more critical.

10 Practice Problems

Recommended exercises from Chapter 5:

5.2, 5.3, 5.5, 5.6, 5.7, 5.14, 5.19, 5.56, 5.57